

Supply Chain Performance Analysis Using Exploratory Data Analysis and Interactive Analytics

Final Research Paper

Abstract

Supply-chain efficiency plays an important role in business performance, profitability, customer value, and the movement of products across markets. This study presents a data-driven analysis of an APL Logistics supply-chain dataset containing more than 180,000 transaction records. The analysis focuses on revenue and profitability, customer value, product and category performance, and the relationship between discounting and profitability.

The study applies data cleaning, exploratory data analysis (EDA), statistical analysis, and interactive visualization. An interactive Streamlit application was developed to allow users to upload the cleaned dataset and explore revenue and profit performance, customer contribution, product and category profitability, and discount-related scenarios. The analysis identifies areas of strong and weak commercial performance and provides a structured basis for managerial recommendations.

The project demonstrates how exploratory data analysis and interactive dashboards can transform a large transactional dataset into an accessible decision-support tool for supply-chain and business analysis.

Keywords

Supply Chain Analytics; Logistics; Exploratory Data Analysis; Revenue; Profitability; Customer Value; Product Performance; Discount Analysis; Streamlit; Data Visualization.

1. Introduction

Supply-chain management requires effective coordination of products, customers, markets, logistics activities, and commercial decisions. The increasing availability of transactional data provides organizations with an opportunity to evaluate these activities using analytical methods rather than relying only on conventional reports.

This project analyzes an APL Logistics supply-chain dataset using exploratory data analysis and interactive visualization. The analysis examines financial and commercial performance across markets, customers, products, categories, and discount levels. A Streamlit dashboard was developed to make the analysis accessible and interactive.

2. Problem Statement

Large supply-chain datasets contain information about customers, products, categories, sales, profit, discounts, markets, regions, and logistics operations. However, raw transactional data does not by itself provide an immediate understanding of business performance.

The study therefore addresses the following question:

How can exploratory data analysis and interactive analytics be used to identify important patterns in supply-chain revenue, profitability, customer value, product performance, and discounting?

3. Objectives

1. Examine overall revenue and profitability.
2. Analyze customer and customer-segment contribution.
3. Identify high- and low-performing products and categories.
4. Examine product-level and category-level profit margins.
5. Investigate the relationship between discounts and profitability.
6. Identify areas requiring managerial attention.
7. Develop an interactive Streamlit dashboard.
8. Provide practical recommendations based on the observed patterns.

4. Dataset Description

The project uses a cleaned APL Logistics supply-chain dataset containing more than 180,000 transaction records. The dataset includes information related to customers, customer segments, products, product categories, sales, profit, quantity, discounts, markets, countries, regions, shipping modes, and delivery-risk information.

The dataset was cleaned and validated before analysis. The application also validates required fields when a user uploads the dataset.

5. Methodology

The project follows a structured workflow:

Dataset → Data Cleaning → Data Validation → Exploratory Data Analysis → Business Analysis → Visualization → Insights → Recommendations

Python-based analysis was performed using Pandas and NumPy. Plotly was used for interactive visualizations and Streamlit was used to build the web application.

5.1 Data Cleaning and Validation

The dataset was checked for data-quality issues including missing values, duplicate records, inconsistent column naming, inappropriate data types, and invalid values. Required fields were also validated before analysis.

5.2 Exploratory Data Analysis

EDA included descriptive statistics, aggregation by market and category, customer-level analysis, product-level analysis, profit-margin calculations, discount-band analysis, and correlation analysis.

6. Exploratory Data Analysis and Findings

6.1 Revenue and Profitability

Revenue and profit were examined at overall and market levels. The analysis compares total sales, total profit, profit margin, transaction volume, units sold, and average order value. Market and category comparisons show that revenue and profitability are not necessarily distributed equally. A high-revenue market or category may not have the strongest profit margin, making profitability analysis important alongside sales analysis.

6.2 Customer Value

Customer-level aggregation was used to evaluate revenue, profit, transaction volume, and profit margin. The analysis identifies top and bottom customers by profit and compares customer segments by revenue, profit, and customer count. This provides a clearer view of where commercial value is concentrated.

6.3 Product and Category Performance

Products were evaluated using revenue, profit, quantity, discount, and profit margin. Category performance was examined using a performance matrix that distinguishes high-revenue/high-profit, high-revenue/low-profit, low-revenue/high-profit, and low-revenue/low-profit categories. High-revenue/low-profit areas are particularly useful for further investigation of pricing, discounting, costs, and operational efficiency.

6.4 Discount Impact

Discount levels were examined using discount bands and visual comparisons against profit and profit margin. Correlations were calculated to identify associations between discount levels and profitability. These results must be interpreted carefully: correlation indicates association and does not establish that discounts cause changes in profitability.

7. Key Insights

The EDA provides four broad insights. First, sales volume and profitability should be evaluated together because high revenue does not automatically imply strong margins. Second, customer contribution varies and customer-level profitability provides information beyond transaction volume. Third, product and category performance should be assessed using both commercial scale and profitability. Fourth, discounting should be monitored because discount levels can be associated with different profitability patterns, although the relationship is not necessarily causal.

8. Recommendations

1. Monitor high-revenue, low-profit categories and investigate pricing, discounting, product costs, and operational efficiency.
2. Track high-value customers and customer segments using profitability as well as revenue.
3. Evaluate products using revenue, quantity, discount, profit, and margin rather than sales volume alone.
4. Review discount strategies with profitability in mind instead of focusing only on sales growth.
5. Use interactive dashboards to make supply-chain information easier for business stakeholders to explore and interpret.

9. Streamlit Application

The project includes a Streamlit dashboard with four modules: Revenue & Profit Overview; Customer Value Dashboard; Product & Category Performance; and Discount Impact Analyzer.

The application uses an upload-based workflow. Users upload the cleaned CSV, the application validates the required fields, and the dashboard then performs the analysis. This approach avoids storing the large dataset inside the GitHub repository. Streamlit's file uploader supports direct CSV uploads and stores uploaded files in memory for the application session. ■cite■turn0search0■turn0search6■

The live application is available at: <https://supplychainanalysis.streamlit.app/>

10. Limitations

1. The analysis is based on the variables available in the supplied dataset.
2. The dataset does not contain a suitable order-date field for reliable monthly time-series analysis.
3. Correlation does not establish causation.
4. The discount what-if tool is an analytical scenario estimate rather than a financial forecast.
5. External factors such as fuel costs, macroeconomic conditions, infrastructure conditions, and supplier performance are not included unless represented in the dataset.

11. Conclusion

This project demonstrates how exploratory data analysis and interactive visualization can be used to examine supply-chain performance across revenue, profitability, customers, products, categories, and discounts.

The resulting Streamlit dashboard converts the analysis into an interactive decision-support tool. The research paper and executive analysis provide complementary interpretations for academic and business audiences.

Future work could incorporate reliable time-series information, transportation and cost variables, external economic indicators, and predictive models for specific supply-chain risks.

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